SOLUTIONS: DISCRETE EVENT SYSTEMS
MASTER IN CONTROL

1. Exercise

   a) We denote rooms as 11, 12, 13, 21, 22 and 23 respectively, with 11 the start room and 23 the finish room. The event set is $E = \{e, w, n, s\}$ and the associated transition diagram is shown in Fig. 1.1. [5]

   b) The labelling automaton G_L is shown in Fig. 1.2. The automaton $\tilde{G}_R$ is shown in Fig. 1.3. [2]

   c) The parallel composition $G = \tilde{G}_R || G_L$ is shown in Fig. 1.4. [4]

   d) The non-deterministic automaton G_N is simply obtained from G by replacing m labels with ε labels. [1]

   e) The observer automaton $\text{Obs}(G_N)$ is shown in Fig. 1.5. [5]

   f) The occurrence of mistakes after a sequence w of s and e events can be detected by the Observer automaton by considering the state $x = f_{obs}(\{11N\}, w)$, and seeing if x state is a union of states that do not contain the label N, such as the state $\{13Y, 12Y, 11Y\}$ in the top right corner of its transition diagram.
   From the physical point of view this happens when the sequence of s and e events considered does not correspond to a possible path in the labyrinth, viz. it goes through walls. [3]

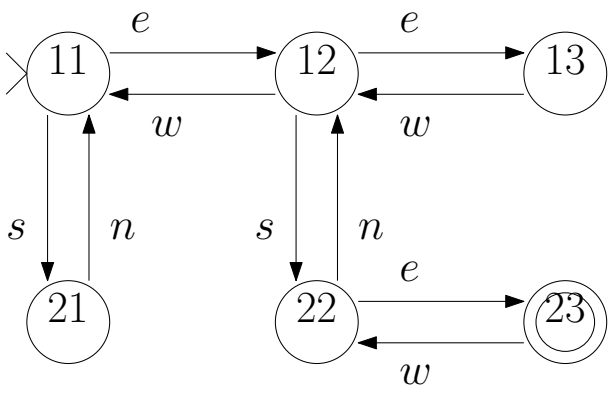


Figure 1.1 Automaton G_R, modelling the robot within the labyrinth

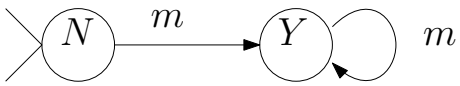


Figure 1.2 Labelling automaton G_L

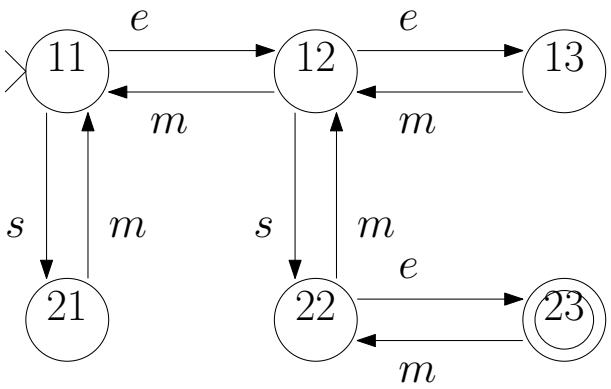


Figure 1.3 Automaton $\tilde{G}_R$, modelling the robot with s, e and m events

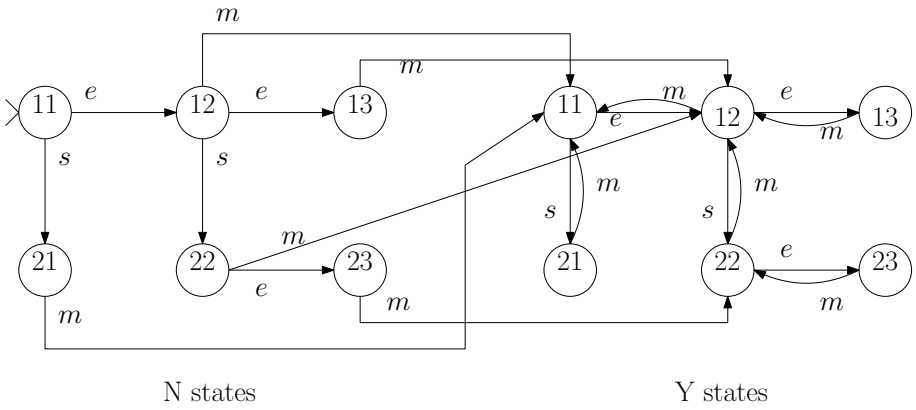


Figure 1.4 Parallel composition $G = \tilde{G}_R || G_L$

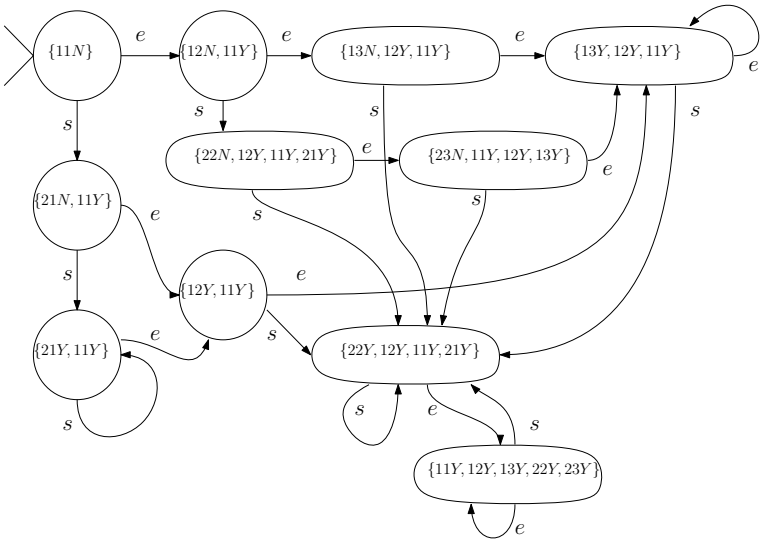


Figure 1.5 Observer automaton Obs(G_N)

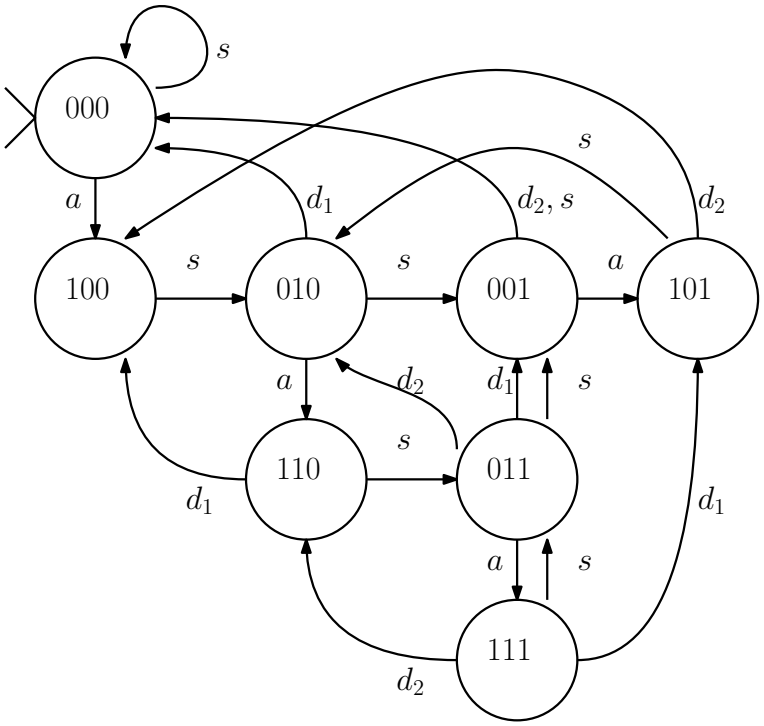


Figure 2.1 Automaton G, modelling the conveyor belt

2. Exercise

a) We model the conveyor belt according to the finite deterministic automaton G, sketched in Fig. 2.1 [6]

b) To avoid pieces falling into the unprocessed box it is enough to remove occurrence of events of type s when the righmost slot in the conveyor is full. This leads to the specification shown in Fig. 2.2. [4]

c) It is clear that $G_S||G \sim G_S$ meaning that the parallel composition automaton has a one to one correspondence with the automaton G_S. In particular, $\Gamma_{G_S||G}((x_s,x)) = \Gamma_{G_S}(Ix_s)$, viz. the same events are enabled in each state of $G_S||G$ as the events enabled in the corresponding state of G_S. In particular, only s events are disabled when carrying out the parallel composition. Since these are controllable events, the specification is controllable. The supervisor is simply given by:

$$S(s) = \Gamma_{G_S}(f(x_0,s)).$$

[4]

d) The specification is uncontrollable if $E_{uc} = \{s\}$. This is because the s event is disabled in several states of the parallel composition. Namely:

$$[\Gamma(x) \backslash \Gamma_{G_s||G}((x_s,x))] \cap E_{uc} \neq \emptyset.$$

This happens for $x = 111, 011, 101, 001$. [2]

e) The maximal controllable sublanguage is unfortunately very small, and equals $\{s\}^*$. In particular, this is generated by the automaton with a single state 000 and the self-loop $f(000,s) = 000$. All arrival events need to be disabled as they may lead to violation of the specification due to the impossibility of preventing shift events and of forcing d_1 and d_2 events. [4]

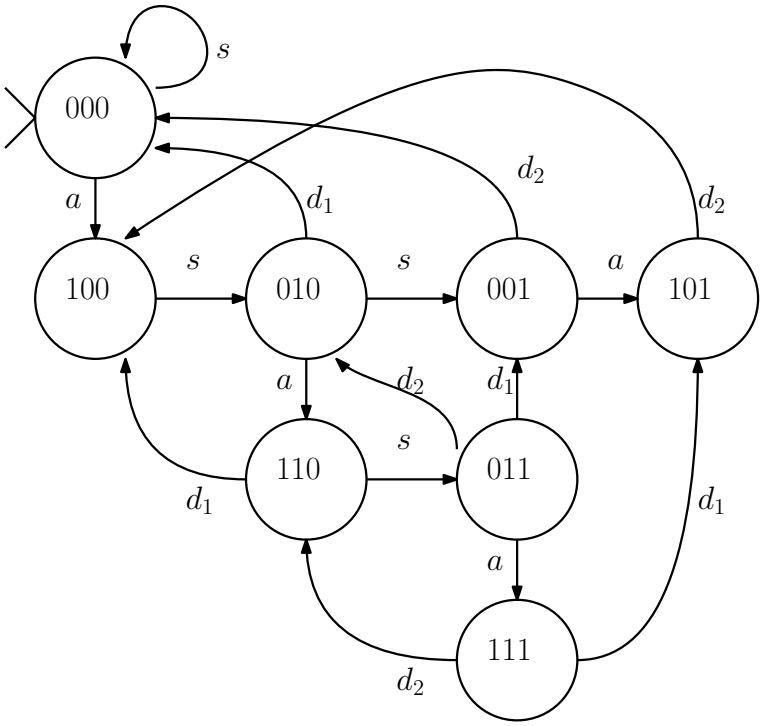


Figure 2.2 Automaton G_S, modelling the specification

3. Exercise

a) The sought matrices are given by:

$$Pre = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \qquad Post = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C = Post - Pre = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}$$

[5]

b) There are 3 minimal P-semiflows in the considered Petri Net. In particular, $[1,1,0,0,0,0,0]$, $[0,0,1,1,0,0,1]$ and $[0,0,0,0,1,1,1]$. [3]
These correspond to: total number of $3d$ printing machines, $[1,1,0,0,0,0,0]$. Total process capacity of upper thread of assembly line, $[0,0,1,1,0,0,1]$. Total process capacity of lower thread of assembly line, $[0,0,0,0,1,1,1]$. [1]

c) Adding up the P-semiflows of minimal support we see that $[1,1,1,1,1,1,2]$ is also a P-semiflow. Its support coincide with P. Hence the Petri Net is structurally (weakly) conservative. [2]

d) There is a single T-semiflow. This is the vector $[1,1,1,1,1]'$. [3]

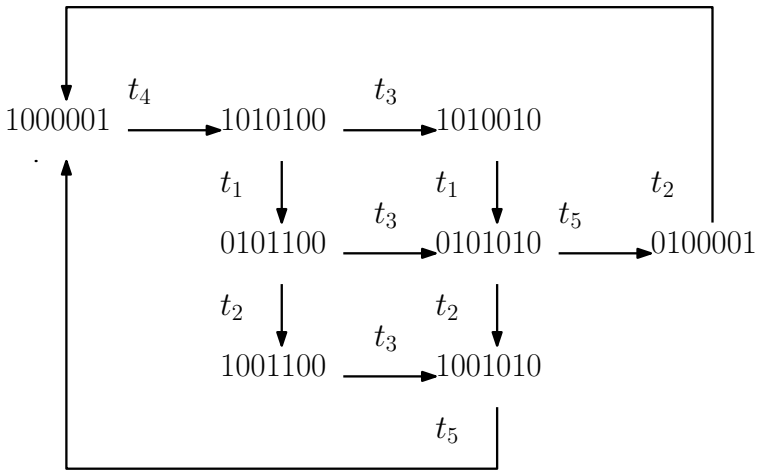


Figure 3.1 Transition diagram of $\mathscr{R}(M_0)$

e) The transition diagram associated to the initial marking M_0 is shown in Fig. 3.1. [4]

f) Notice that the transition diagram is strongly connected. Hence, the Petri Net N, M_0 is reversible. In particular, it is always possible to go back to the initial marking M_0 through a suitable sequence of transitions. [2]

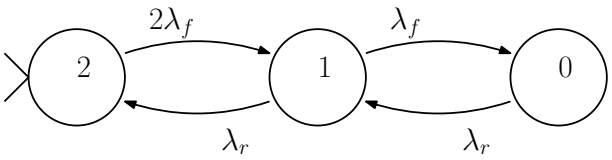


Figure 4.1 Transition diagram of the Markov Chain

4. Exercise

a) We model the factory according to a Markov chain with 3 states, labeled as 2, 1 and 0 in Fig. 4.1, according to the number of machines available in each state. Transition from state 2 to state 1 occurs with rate $2\lambda_f$ as any of the 2 machines may experience a fault. On the other hand, transition from 1 to 0 occurs with rate λ_f as only the operational machine may experience the fault. Conversely, transitions from 1 to 2 and 0 to 1 happen with a rate λ_r, independently of the number of faulty machines as the technician can only work on one of them at a time. [3]

Its Kolmogorov equations are given below:

$$\dot{\pi} = \pi \begin{bmatrix} -2\lambda_f & 2\lambda_f & 0 \\ \lambda_r & -(\lambda_f+\lambda_r) & \lambda_f \\ 0 & \lambda_r & -\lambda_r \end{bmatrix} := \pi Q.$$

[3]

b) This is an ergodic Markov chain as it can be seen due to strong connectedness of its transition diagram (or, equivalently, irreducibility of its associated matrix Q). [2]

c) To compute the average throughput we first compute the steady-state probability distribution, according to the linear system of equations:

$$\pi Q = 0, \qquad \pi \mathbf{1} = 1.$$

Its solution is given as:

$$\pi_\infty = \left[\frac{\lambda_r^2}{2\lambda_f^2+2\lambda_f\lambda_r+\lambda_r^2}, \frac{2\lambda_f\lambda_r}{2\lambda_f^2+2\lambda_f\lambda_r+\lambda_r^2}, \frac{2\lambda_f^2}{2\lambda_f^2+2\lambda_f\lambda_r+\lambda_r^2} \right].$$

[3]

By the ergodic theorem, the average throughput is computed as:

$$[2K, K, 0]\pi_\infty^T = K\frac{2\lambda_r^2+2\lambda_f\lambda_r}{2\lambda_f^2+2\lambda_f\lambda_r+\lambda_r^2}.$$

[2]

d) Differentiating the expression of the average asymptotic throughput with respect to λ_r yields:

$$\frac{d}{d\lambda_r} K\frac{2\lambda_r^2+2\lambda_f\lambda_r}{2\lambda_f^2+2\lambda_f\lambda_r+\lambda_r^2} = K\frac{2\lambda_f(2\lambda_f^2+4\lambda_f\lambda_r+\lambda_r^2)}{(2\lambda_f^2+2\lambda_f\lambda_r+\lambda_r^2)^2}.$$

This is clearly positive, regardless of λ_r, λ_f or K. Hence increasing λ_r improves the throughput. [2]

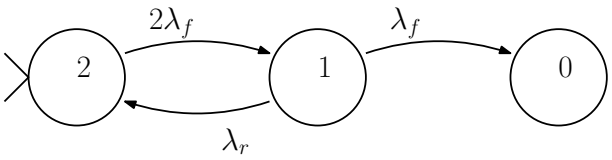


Figure 4.2 Modified Markov Chain for computation of absorption time

e) We first modify the Markov Chain as in Fig. 4.2. [2]

We then compute the Laplace transform of $\dot{\pi}_3$, for initial condition $\pi(0) = [1,0,0]$ which is given by:

$$\mathscr{L}(\dot{\pi}_3) = \frac{2\lambda_f^2}{s^2 + (\lambda_r + 3\lambda_f)s + 2\lambda_f^2)}.$$

Finally the average absorption time is given by:

$$\tau_a = -\frac{d}{ds}\mathscr{L}(\dot{\pi}_3)\Big|_{s=0} = \frac{3\lambda_f + \lambda_r}{2\lambda_f^2}.$$

[3]